

Diffusion modeling for predicting Functional Molecule Conformers

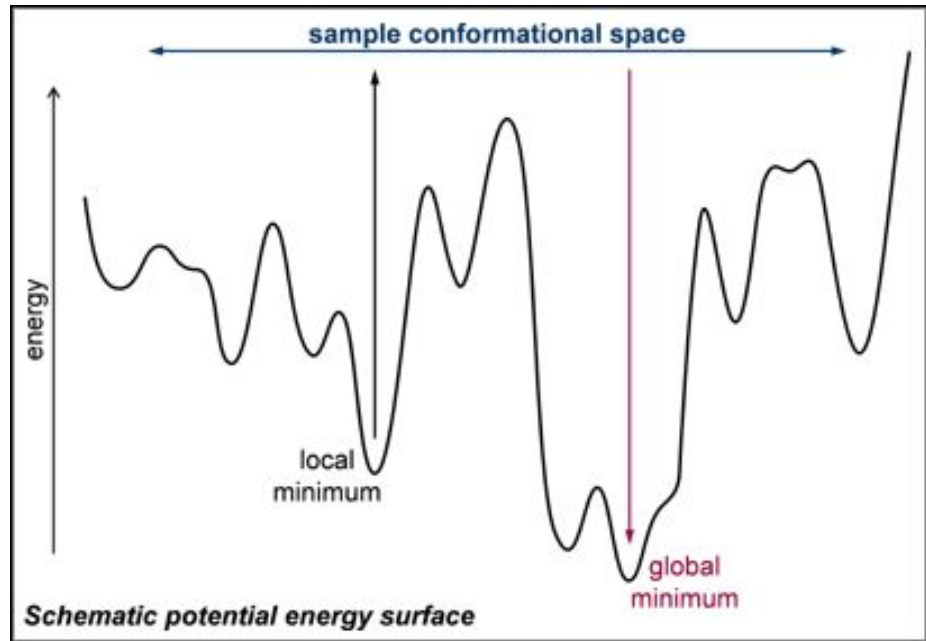
Team members

Sudheesh Kumar Ethirajan

Saurabh Sivakumar

Molecular Conformers & Their Importance

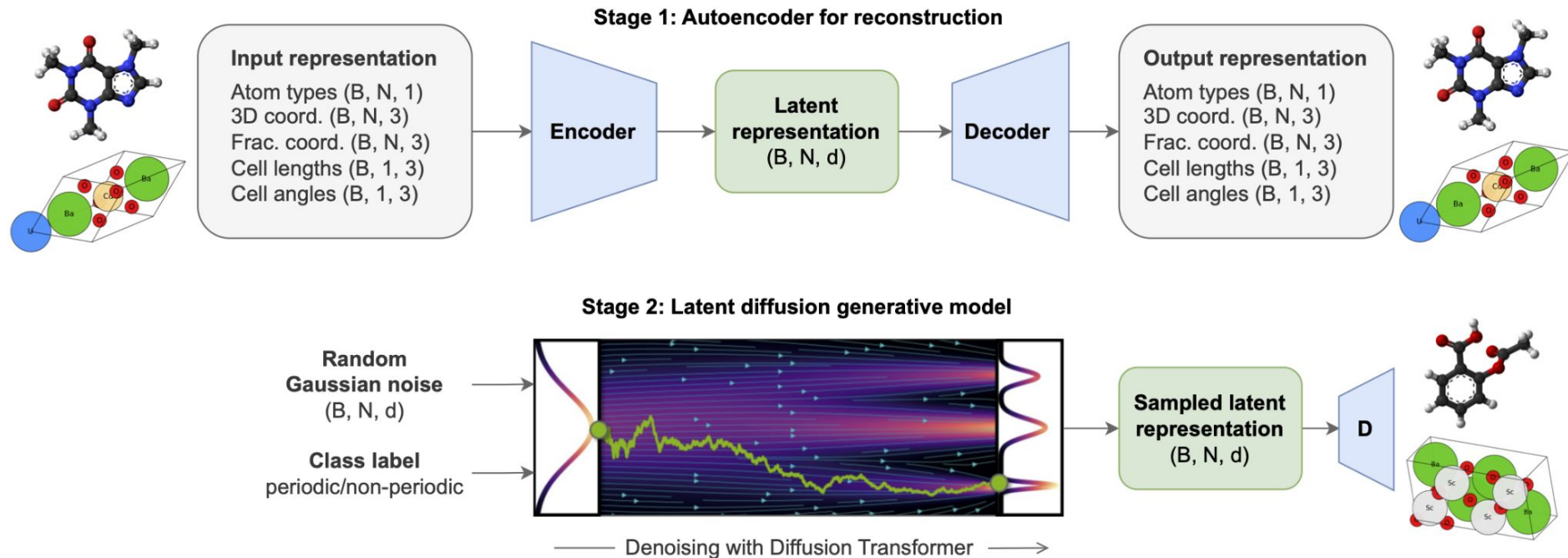
- **Molecular Conformers:** Spatial arrangements of atoms resulting from rotations about single bonds.
- Relevance to drug discovery, catalysis, and materials modeling.
- Traditional methods: Accurate but expensive.
- Challenge: Sampling diverse, low-energy conformations efficiently.



Why Generative Modeling?

- Probabilistic generators can learn the full distribution of valid 3-D conformers.
- **Why diffusion models?**
 - Score-based diffusion allows fine-grained control and shows state-of-the-art performance on structured 3-D data, training directly on existing ensembles.
- **Practical aim:** Replace or shorten expensive existing methods with fast generative inference.
- **Energy-Aware Diffusion (ADiT):** Extends diffusion by conditioning on potential energy, not just geometry.
- **Outcome:** Guides generation toward low-energy, chemically realistic conformers, improving structure-property fidelity for design and simulation tasks.

ADiT — Two-Stage Generative Framework



Stage 1 — Autoencoder Latent Representation

- **Encoder-** Based on SE(3)-equivariant Transformers
- **Latent Representation-** Preserve molecular symmetries and encode atomic environments.
- **Decoder-** Transformer
 - Atom types via classification.
 - Fractional coordinates and cell vectors for periodic structures.
 - Cartesian coordinates and distances for non-periodic molecules (e.g., QM9).
- **Loss Functions**
 - Reconstruction losses:
 - Atom type classification (cross-entropy)
 - Position regression (MSE)
 - KL divergence- to regularize the latent distribution for generative use.
- **Key Advantages**
 - Enables multi-domain learning in a unified latent space.
 - Preserves symmetry-awareness using equivariant modeling.
 - Produces a latent space well-suited for diffusion-based generation in Stage 2.

Stage 2 — Latent Diffusion with Flow Matching

- **DiT Denoiser (Diffusion Transformer)**
 - Inputs: noisy latents x_t , timestep t , optional conditioning (e.g. dataset index, space group).
 - Outputs: prediction of clean latent (x_0)
- **Interpolant (Flow Matching)**- Uses Flow Matching instead of the traditional score-based
 - Learns the velocity field to match the true data flow from noise to clean samples.
 - Supervised training using known interpolation pairs (x_t, x_0) ,
- **Training Process**
 - Randomly sample a timestep $t \in [0, 1]$
 - Generate noisy latent x_t via linear interpolation, $x_t = (1 - t) \cdot x_0 + t \cdot x_1$
 - Predict velocity v_{pred} using DiT
 - Supervise with ground truth velocity, $v = x_1 - x_0$
 - Loss is the MSE between v and v_{pred}
- **Advantages of This Setup**
 - Works in latent space → Reduces complexity vs. direct coordinate modeling
 - Flow Matching → More stable & efficient training than score-based diffusion
 - Transformer-based Denoiser (DiT) → Captures long-range dependencies between atoms

Energy-Aware Extension

- **Motivation**

- Including energy helps guide the model toward energetically favorable
- Ensures better structure-property alignment for downstream tasks (e.g., molecular design)

- **Key Design Choices**

- Energy Prediction Head
 - Added a small 2-layer MLP that maps the latent vector to a scalar energy value.
 - Input: $Z_{\text{mean}} \in \mathbb{R}^D$, the mean pooled latent representation
 - Output: Predicted internal energy at 0 K
- Loss Function
 - Introduced energy-aware loss as a regularization term, $\mathbf{L}_{\text{total}} = \mathbf{L}_{\text{flow}} + \lambda_{\text{energy}} \cdot \mathbf{L}_{\text{energy}}$
- Training Strategy
 - Only enabled energy loss when energy labels (from QM9) are available.
 - During training: frozen VAE, flow-matching denoiser learns to minimize both denoising and energy prediction errors.

Experiments Conducted

- **Dataset Setup**

- QM9 contains ~130k small organic molecules with 3D conformers and quantum properties.
- Energy at 0 K used for energy-aware training.
- Data split: 80%, 10%, 10% (train, val, test)

- **Training Pipeline**

- Stage 1: Used the checkpoint for VAE shared by Meta
- Stage 2: Fine-tune latent diffusion model using Flow Matching
 - $\lambda_{\text{energy}} = 0$
 - $\lambda_{\text{energy}} = 0.01$
 - $\lambda_{\text{energy}} = 0.1$
 - $\lambda_{\text{energy}} = 0.2$

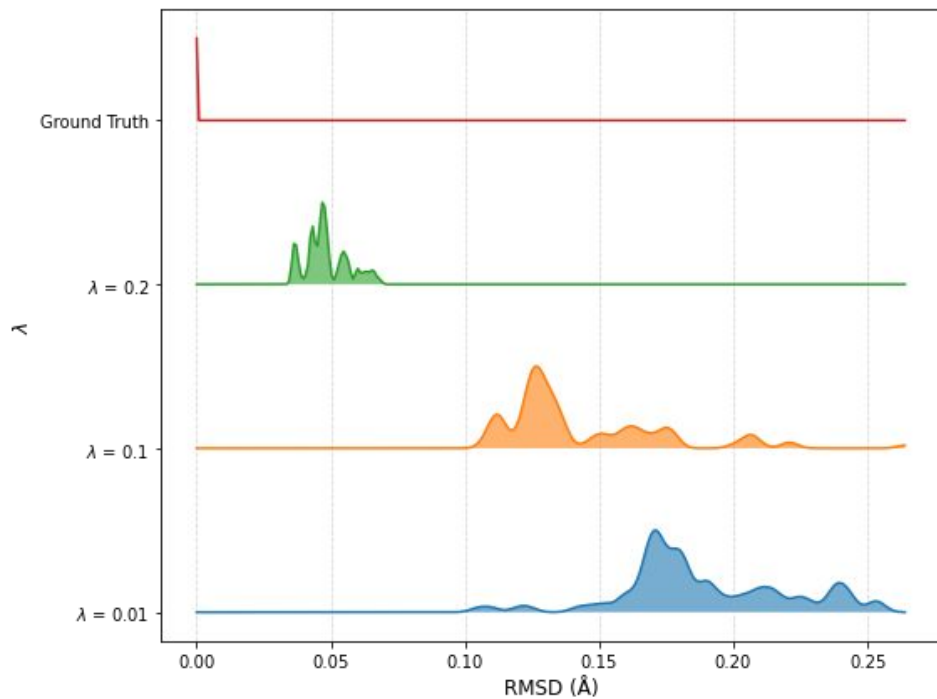
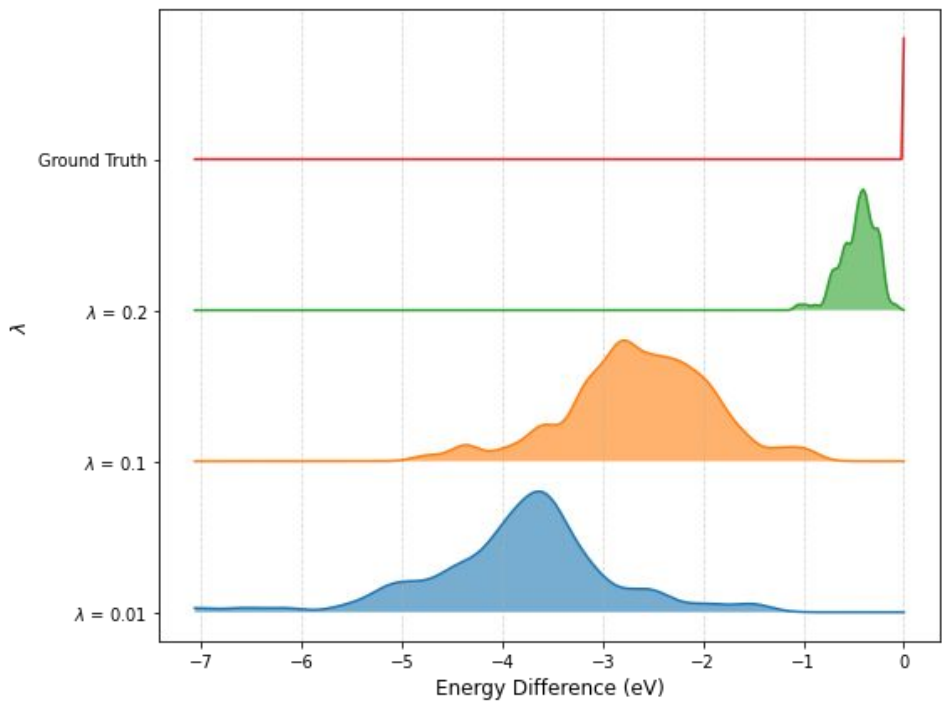
- **Sampling Procedure**

- Used Classifier-Free Guidance (CFG) during generation.
- Sampled 100 molecules per evaluation round.
- Evaluated the ground truth energy and optimized structure of the conformers

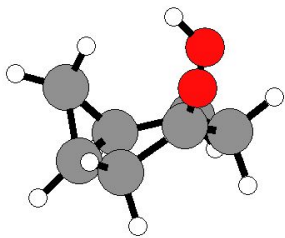
Model Fine-Tuning



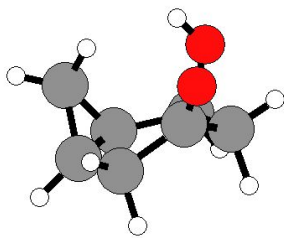
Distributions of Sampled Conformers



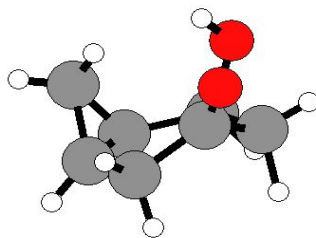
Conformer visualizations



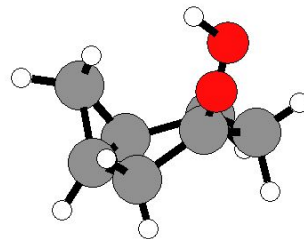
Ground Truth



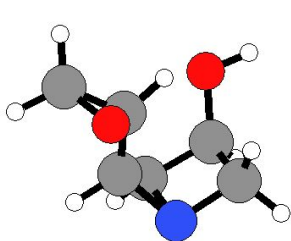
$\lambda = 0.2$



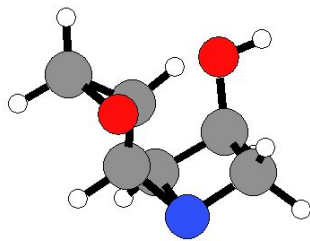
$\lambda = 0.1$



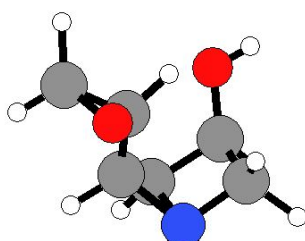
$\lambda = 0.01$



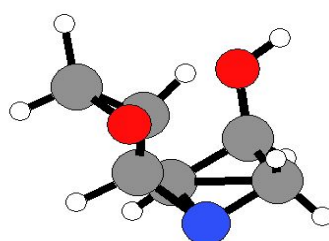
Ground Truth



$\lambda = 0.2$



$\lambda = 0.1$



$\lambda = 0.01$

Conclusion & Future Work

- **Conclusions**

- ADiT introduces a modular two-stage pipeline combining variational autoencoding and latent-space diffusion to generate 3D molecular conformers.
- Energy-aware training improves physical plausibility by penalizing deviation from ground truth internal energy.
- Empirical evaluations on QM9 show strong validity, diversity, and energetic consistency of sampled molecules.

- **Future Work**

- Extend to periodic crystals like MOFs and inorganic frameworks leveraging spacegroup conditioning.
- Compare flow matching with other diffusion solvers (e.g., probability flow ODE, Schrödinger bridge)

